

Muhammad Umar Farooq

✉ muf@umich.edu

📍 Ann Arbor, Michigan, USA

🌐 iammufarooq

🔗 dr-muhammadumarfarooq

🆔 0000-0003-4139-2082

Education

University of Michigan, Ann Arbor , Mechanical Engineering	Ann Arbor, Michigan, USA
Dissertation: From Process to Supply Chain: Efficient AI for Manufacturing Decarbonization	2023 – 2026
University of Leeds and TRIBOS+ Consortium Europe , Tribology of Surfaces and Interfaces	Europe
- Institutions: University of Leeds, University of Ljubljana, University of Coimbra, and Luleå University of Technology - Erasmus Mundus Joint Master Degree scholarship recipient - Thesis: Tribological Performance of Ball Bearing Subjected to an Electric Field	2021 – 2023
University of Engineering and Technology Lahore , Engineering Management	Lahore, Pakistan
- Fully funded master degree scholarship recipient - Thesis: Assessing Technology (Industry 4.0 Transformation) Readiness of Industries to Combat Future COVID-19 like Disruptions: A Case of Pakistan's Manufacturing Sector	2019 – 2021
University of Engineering and Technology Lahore , Industrial and Manufacturing Engineering	Lahore, Pakistan
- President's Gold Medal - Fully funded degree scholarship recipient - Thesis: Parametric optimization of powder mixed electric discharge machining process for aeronautical alloys and composites	2015 – 2019

Experience

Ford Motor Company , Research Lead, Sheet Metal Components: Process to Supply Chain Project	Michigan, USA
- Lead research connecting sheet-metal manufacturing processes with supply-chain analysis and engineering decision support - Develop data-driven sustainability and carbon-accounting methods for automotive manufacturing applications	Nov 2024 – present 1 year 9 months
Resourceful Manufacturing and Design (ReMaDe) Lab, University of Michigan , Graduate Student Research Assistant	Ann Arbor, Michigan, USA
- Lead technical contributor on physics-integrated AI and carbon-accounting tools for manufacturing and supply chains - Develop reproducible workflows for validation, uncertainty reporting, and engineering decision support - Mentor and onboard new researchers in Python, version control, and research documentation	Aug 2023 – present 3 years 1 month
The Sargent Centre for Process Systems Engineering, University College London , Visiting Researcher	London, United Kingdom
Collaborated on explainable machine-learning methods for modeling and optimization under uncertainty in engineered systems	Sept 2024 – Sept 2024 1 month
Scania AB , Research Student, TRIBOS+ Industry Project	Södertälje, Sweden
Studied tribological behavior of machine elements under electrified conditions with applications to electric-vehicle drivetrain durability	Sept 2022 – Aug 2023 1 year

SKF Research Center , Software Engineering Intern Developed an automated signal-processing tool with feature engineering for bearing diagnostics in high-speed rail applications	Nieuwegein, Netherlands July 2022 – Aug 2022 2 months
Lahore University of Management Sciences , Research Associate Contributed to empirical research on project leadership, external environmental factors, and project success	Lahore, Pakistan Mar 2021 – Aug 2021 6 months
University of Engineering and Technology Lahore , Graduate Teaching Assistant and Laboratory Instructor - Led Manufacturing Processes Laboratory instruction for more than 40 students and developed laboratory activities and assessment rubrics - Supported Industrial Materials and Production Planning and Control courses	Lahore, Pakistan Nov 2019 – Oct 2020 1 year

Awards

Rackham Conference Travel Award US\$1,400	University of Michigan 2026
Rackham Conference Travel Award US\$1,150	University of Michigan 2025
Outstanding Reviewer Award	Manufacturing Letters (SME) 2024
Rackham Conference Travel Award US\$900	University of Michigan 2024
Rackham Graduate Student Research Grant US\$1,500	University of Michigan 2024
Erasmus Mundus Scholarship EUR 47,000	European Commission 2021
President's Gold Medal	UET Lahore 2019

Certificates

Oxford Machine Learning Summer School University of Oxford	July 2024
2D/3D Forming Process Modelling Scientific Forming Technologies Corporation	Dec 2023
McKinsey Forward Program McKinsey & Company	Dec 2022
7th International Deep Learning School Luleå University of Technology	Oct 2022

Skills

Research Areas: Sustainable manufacturing, physics-informed and data-driven modeling, multi-fidelity learning, uncertainty quantification, optimization under uncertainty, life cycle assessment, supply chain decarbonization, and intelligent manufacturing systems

Programming and Software: Python, scientific computing, machine learning, optimization, reproducible workflows, and Git